MIL-P-43081B(MR)
2 November 1973
SUPERSEDING
MIL-P-43081A(MR)
29 January 1968

MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL (POTTING AND IMPREGNATING), POLYETHYLENE, LOW MOLECULAR WEIGHT

1. SCOPE

1.1 Scope. This specification covers two classes of low molecular weight polyethylene material.

1.2 Classification. The low molecular weight resins shall be of the following classes, as specified (see 6.2).

Class 1 - General purpose polyethylene without antioxidant
Class 2 - General purpose polyethylene with antioxidant

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attribute
MIL-STD-129 - Marking for Shipment and Storage

FSC 9330

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D36-70	Softening Point of Asphalts and Tar Pitches (Ring-and-Ball Apparatus).
D150-70	A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.
D374-68	Thickness of Solid Electrical Insulation.
D618-61 (1971)	Conditioning Plastics and Electrical Insulating Materials for Testing.
D792-66 (1970)	Specific Gravity and Density of Plastics by Displacement.
D1238-70	Flow Rates of Thermoplastics by Extrusion Plastometer.
D1505-68	Density of Plastics by the Density-Gradient Technique.
D1531-62 (1970)	Dielectric Constant and Dissipation Factor of Polyethylene by Liquid Displacement Procedure.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

National Motor Freight Traffic Association, Inc., Agent:

Nation Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 First article sample. When specified by the procuring activity (see 6.2), the material furnished under this specification shall be a product which has been subjected to and has passed the first article sample inspection specified herein (see 4.2.3 and 4.2.3.1).

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3.2 Material. The material shall consist of low molecular weight polyethylene resin (see 6.3) in the granular form. Class 1 Material shall be without antioxidant, and class 2 material shall contain antioxidant.

3.3 Property values. The material shall conform to the property values specified in Table I, when tested as specified in the applicable procedure of 4.3.

Table I. Property Values

Property	Value			
	Class 1		Class 2	
	Minimum	Maximum	Minimum	Maximum
Flow rate[1]/ Condition A (grams per 10 minutes at 125°C (266°F) and 6.5 psi)	6	11	-	-
Condition E, melt index, (grams per 10 minutes at 190°C (374°F) and 43.25 psi)	300	500	170	230
Density, g/cm^3, (specific gravity)	0.850	0.915	0.850	0.915
Softening point, (glycerine) degrees Centigrade	104	116	104	116
degrees Fahrenheit	220	240	220	240
Dielectric constant at 1 MH_z[2]/	2.20	2.35	2.20	2.35
Dissipation factor at 1 MH_z[2]/ D-tan delta	-	0.0005	-	0.0005

[1]/For class 1 material, flow rate shall be determined either at condition A or condition E of ASTM D1238-70 (see 4.3.3), at the option of the supplier.

[2]/1 MH_z[2]/ equals 1 megacycle per second.

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3.4 Color. Unless otherwise specified by the procuring activity, (see 6.2), material granules shall be natural in color.

3.5 Uniformity. All material shall be uniform in color and form as determined visually. The form shall be granules.

3.6 Workmanship. The material shall be clean and free of foreign particles and other contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification will govern. A supplier's certificate of compliance with 3.2 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 50 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

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4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table II shall be approximately one pound.

Table II. Examination of the material for defects in appearance and workmanship.

Examine	Defect
Appearance and Workmanship	Color not as specified Not translucent, when specified Not opaque when specified Form or color not uniform Form not as specified Not clean, presence of foreign material

4.2.2.1.2 Examination of the preparation for delivery. An examination shall be made in accordance with table III, to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table III. Examination of preparation for delivery

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers
Quantity of Material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Classification of inspection. All inspection shall be classified as follows:

(a) First article sample inspection (see 4.2.3.1).

(b) Lot acceptance inspection (see 4.2.3.2).

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4.2.3.1 First article sample inspection. When a first article sample is required (see 3.1), it shall be inspected to determine conformance to all requirements of this specification.

4.2.3.2 Lot acceptance inspection. Each lot of material shall be inspected to determine conformance to all requirements of this specification and shall serve as a basis for acceptance or rejection of the lot.

4.2.4 Testing. Each lot of molding material shall be tested for the applicable characteristics listed in Table I, in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 50 pounds. The sample unit shall consist of sufficient material to prepare all required specimens, and shall be a thoroughly composited mixture of 5 approximately equal units randomly selected from the lot. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be the averaged results of the specimens.

4.3 Test methods.

4.3.1 Specimen preparation. The material in the form of granules shall be used for flow rate testing. Specimens shall be prepared for the other testing by molding in accordance with manufacturer's instructions.

4.3.2 Specimen conditioning. Except for softening point determination, molded test specimens and granules as applicable, shall be conditioned in accordance with procedure A of ASTM D618-61 (1971). Testing shall be at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity.

4.3.3 Flow rate. Flow rate shall be determined in accordance with ASTM D1238-70. Three determinations shall be made. Either procedure A or B, and either condition A or condition E, as applicable, or ASTM D1238-70 shall be used.

4.3.4 Density. Two specimens shall be tested in accordance with either ASTM D1505-68, or ASTM D792-66(1970) using method A-1 or method A-2.

4.3.5 Softening point. Two specimens shall be tested in accordance with ASTM D36-70.

4.3.6 Dielectric constant and dissipation factor. Three specimens shall be tested in accordance with either ASTM D1531-62(1970) or ASTM D150-70, with ASTM D1531-62(1970) preferred as the more precise method. When ASTM D1531-62(1970) is used, specimens shall be prepared as specified in this method. When ASTM D150-70 is used, specimens shall be 0.125 ± 0.010 inch thick measured in accordance with method A or C of ASTM D374-68. Either foil or silver painted electrodes shall be used with ASTM D150-70.

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5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds, maximum.

(b) Metal drums conforming to PPP-D-729, type III, or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. Packing shall be in a manner to insure acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The low molecular weight polythylene is intended primarily for use in potting and impregnating compounds.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class of material required (see 1.2).

(c) First article sample, if required (see 3.1).

(d) Color, if required (see 3.4).

(e) Selection of applicable level of packing (see 5.1).

6.3 Low molecular weight polythylene. Low molecular weight material covered herein is defined as polyethylene with a number average molecular weight range between 10,000 and 20,000 as obtained by standard industry gel permeation chromatography determinations.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - EL, ME

Project No. 9330-A419

User activities:
Army - MD, MU, AT, DO

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor who use in the preparation of procurements of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-P-43081B(MR), Plastic Molding Material (Potting and Impregnating), Polyethylene, Low Molecular Weight

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS: *Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM (APR 62) 1426 REPLACES NAVSHIPS FORM 4853, WHICH IS OBSOLETE

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